- **The Bioinformatics:** they will handle genomic preprocessing (TPM normalization, batch correction with ComBat) and feature selection of the top-k variable genes.

- **The ML/Architecture:** Responsible for building the Genomic Subnetwork. They will implement the 1D-CNN or Transformer encoder and scale the Gradient Blending formula to handle three modalities.

- **The Data/Clinical:** Focused on the SarcomaAI Playbook tasks. They will manage the AWS/XNAT storage, patient ID mapping between MRI and genomic files, and ensure compliance with REB approvals.

2. Implement the Genomic Preprocessing Pipeline

Your team needs to move genomic data from raw formats to "AI-ready" vectors. Update the SarcomaAI Playbook to include:

- **Standardization:** Convert RNA-seq read counts into Transcripts Per Million (TPM) or FPKM-UQ.

- **Normalization:** Apply a $Log2(x+1)$ transformation to stabilize variance across samples.

- **Dimensionality Reduction:** Use an autoencoder or pathway-aware masking (grouping genes into biological functions like "Cell Cycle Control") to reduce $20,000+$ features into a 128-dimension latent vector that matches the current clinical/image embeddings.

3. Supplement the Model Architecture

The goal is to transition the current model from a 2-branch to a 3-branch system.

- **The 3rd Branch:** Add a Genomic Subnetwork as a parallel encoder.
- **Intermediate Fusion:** Instead of just concatenating at the very end, explore "Cross-Modal Attention." This allows the genomic data to "tell" the image network which parts of the MRI (e.g., necrotic areas) are most salient based on molecular aggression.

- **Update the Loss Function:** You must scale the Gradient Blending formula to account for the new branch to prevent the model from ignoring the high-dimensional genomic data:
$$L_{total} = w_{MM}L_{MM} + w_{img}L_{img} + w_{clin}L_{clin} + w_{gen}L_{gen}$$

4. Strategic Data Sourcing (TCGA-SARC)

Because rare disease data is limited, your first "wet run" should use the **TCGA-SARC** dataset. It contains matched MRI/CT images and multi-omic profiles for 206 patients.

- Use TCGA to **pre-train** your genomic encoder.
- Use Dr. Bozzo's internal McGill/MSKCC data to **fine-tune** and validate the model.

5. Documenting for Med School Applications

To make this stand out on your application, you must go beyond "I did research." Focus on:

- **Interdisciplinary Leadership:** Describe how you coordinated between computer scientists and oncologists to build a "bridge" between different data languages.
- **Quantifiable Impact:** Aim to report how much the "C-Index" (predictive accuracy) improved once you added the genomic branch compared to the original image+clinical model.
- **The "Precision" Narrative:** Frame your work as a move away from "median-based" medicine toward individualized, biologically-grounded risk assessment—a core value of modern medical education.